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Unit -3

Q1.Write a program to accept elements of integer, float, and character arrays from the user and display the value and corresponding memory address of each array element.

```
#include <stdio.h>

int main()
{
    int i;
    int a[3];
    float b[3];
    char c[3];

    printf("Enter 3 integer elements:\n");
    for (i = 0; i < 3; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("\nEnter 3 float elements:\n");
    for (i = 0; i < 3; i++)
    {
        scanf("%f", &b[i]);
    }

    printf("\nEnter 3 character elements:\n");
    for (i = 0; i < 3; i++)
    {
        scanf(" %c", &c[i]);
    }

    printf("\nInteger Array:\n");
    for (i = 0; i < 3; i++)
    {
```

```
        printf("Value = %d, Address = %p\n",
                a[i], (void *)&a[i]);
    }

    printf("\nFloat Array:\n");
    for (i = 0; i < 3; i++)
    {
        printf("Value = %.2f, Address = %p\n",
                b[i], (void *)&b[i]);
    }

    printf("\nCharacter Array:\n");
    for (i = 0; i < 3; i++)
    {
        printf("Value = %c, Address = %p\n",
                c[i], (void *)&c[i]);
    }

    return 0;
}
```

OUTPUT-

```
Enter 3 integer elements:
10 20 30

Enter 3 float elements:
1.5 2.5 3.5

Enter 3 character elements:
A B C

Integer Array:
Value = 10, Address = 0x1000
Value = 20, Address = 0x1004
Value = 30, Address = 0x1008

Float Array:
Value = 1.50, Address = 0x2000
Value = 2.50, Address = 0x2004
Value = 3.50, Address = 0x2008

Character Array:
Value = A, Address = 0x3000
Value = B, Address = 0x3001
Value = C, Address = 0x3002
```

Q2. Write a program to accept the elements of a one-dimensional array and calculate the sum of all its elements.

```
#include <stdio.h>

int main()
{
   int a[100], n, i, sum = 0;

   printf("Enter the number of elements: ");
   scanf("%d", &n);

   printf("Enter the array elements:\n");
   for (i = 0; i < n; i++)
   {
      scanf("%d", &a[i]);
   }

   for (i = 0; i < n; i++)
   {
      sum = sum + a[i];
   }

   printf("Sum of array elements = %d\n", sum);

   return 0;
}
```

OUTPUT-

```
Enter the number of elements: 5
Enter the array elements:
10 20 30 40 50
Sum of array elements = 150
```

Q3.Write a program to read two matrices of the same order, perform matrix addition, and display the resultant matrix using a two-dimensional array.

```
#include <stdio.h>

int main()
{
    int a[10][10], b[10][10], sum[10][10];
    int i, j, r, c;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &r, &c);

    printf("Enter the elements of first matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter the elements of second matrix:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            sum[i][j] = a[i][j] + b[i][j];
        }
    }
```

```
    printf("Resultant matrix after addition:\n");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            printf("%d\t", sum[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT-

```
Enter the number of rows and columns: 2 2
Enter the elements of first matrix:
1 2
3 4
Enter the elements of second matrix:
5 6
7 8
Resultant matrix after addition:
6       8
10      12
```

Q4.Write a program to find the factorial of a given number using a user-defined function.

```
#include <stdio.h>

long long factorial(int n)
{
    long long fact = 1;
    int i;

    for (i = 1; i <= n; i++)
    {
        fact = fact * i;
    }

    return fact;
}

int main()
{
    int n;
    long long result;

    printf("Enter a non-negative integer: ");
    scanf("%d", &n);

    if (n < 0 || n > 20)
    {
        printf("Please enter a number between 0 and 20.\n");
    }
    else
    {
        result = factorial(n);
        printf("Factorial of %d = %lld\n", n, result);
    }

    return 0;
}
```

OUTPUT-

```
Enter a non-negative integer: 5
Factorial of 5 = 120
```

Q5.Write a program to calculate the area of a triangle by implementing the following four types of userdefined functions: 5.1) No Arguments, No Return Value 5.2) Arguments, No Return Value 5.3) No Arguments, Returns a Value 5.4) Arguments, Returns a Value

5.1

```
#include <stdio.h>

void area()
{
    float b, h, a;

    printf("Enter base and height: ");
    scanf("%f %f", &b, &h);

    a = 0.5 * b * h;

    printf("Area of triangle = %.2f\n", a);
}

int main()
{
    area();
    return 0;
}
```

OUTPUT-

```
Enter base and height: 10 5
Area of triangle = 25.00
```

5.2

```
#include <stdio.h>

void area(float b, float h)
{
    float a = 0.5 * b * h;
    printf("Area of triangle = %.2f\n", a);
```

```
}

int main()
{
    float b, h;

    printf("Enter base and height: ");
    scanf("%f %f", &b, &h);

    area(b, h);

    return 0;
}
```

OUTPUT-
Enter base and height: 10 5
Area of triangle = 25.00

5.3

```
#include <stdio.h>

float area()
{
    float b, h;

    printf("Enter base and height: ");
    scanf("%f %f", &b, &h);

    return 0.5 * b * h;
}

int main()
{
    float a;

    a = area();

    printf("Area of triangle = %.2f\n", a);

    return 0;
}
```

OUTPUT-
Enter base and height: 10 5

Area of triangle = 25.00

5.4

```
#include <stdio.h>

float area(float b, float h)
{
    return 0.5 * b * h;
}

int main()
{
    float b, h, a;

    printf("Enter base and height: ");
    scanf("%f %f", &b, &h);

    a = area(b, h);

    printf("Area of triangle = %.2f\n", a);

    return 0;
}
```

OUTPUT-

```
Enter base and height: 10 5
Area of triangle = 25.00
```